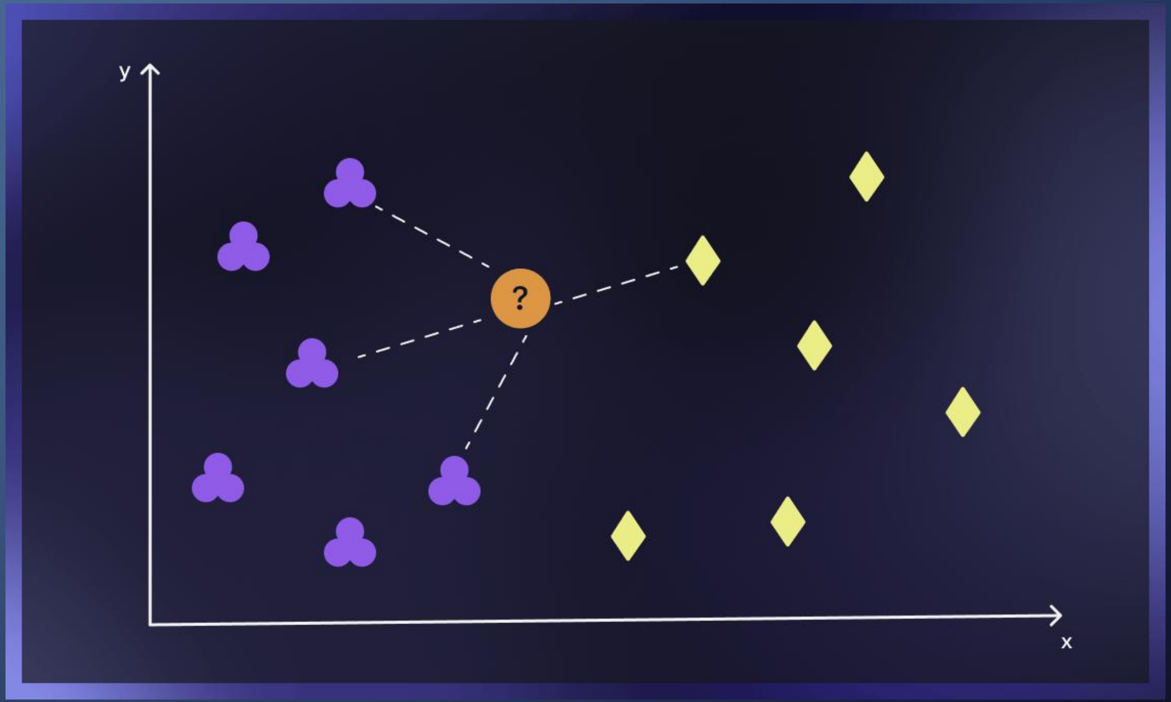
KNN CLASSIFIER FOR NETWORK INTRUSION DETECTION



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INTRODUCTION

- Machine learning is now widely used to identify anomalies, attacks, and unusual traffic patterns in cybersecurity.
- ML models like K-Nearest Neighbor (KNN) are becoming important tools to improve intrusion detection.
- The K-Nearest Neighbor (KNN) classifier can be used on a KDD-based intrusion detection dataset.
- The process of KKN classifier includes preprocessing, classification, confusion matrix evaluation, and decision boundary visualization.
- A contour plot is created to visualize the decision boundary of the KNN classifier.

MACHINE LEARNING (ML) SYSTEMS PLACEMENT

- In the Security Operations Center (SOC) as part of threat detection pipelines.
- Inside the Intrusion Detection/Prevention System (IDS/IPS) to classify network flows in real-time.
- Integrated with SIEM platforms (Splunk, QRadar, Azure Sentinel) to enhance alert accuracy.
- Positioned at network control points gateway routers, firewalls, load balancers.
- Run at endpoint security layers for malware/intrusion behavior analysis.

DIFFERENT ATTACKS AGAINST ML SYSTEMS

- **Evasion Attacks** - Attacker manipulates input so the model misclassifies traffic (slightly modifying attack packets).
- **Poisoning Attacks** - Injecting malicious samples into training data to influence model behavior.
- **Model Extraction Attacks** - Stealing model outputs to recreate or reverse-engineer the classifier.
- **Adversarial Samples -** Crafting traffic patterns that trick ML systems by exploiting algorithmic weaknesses.

DATASET DESCRIPTION

- KDD-based dataset (lab_data_kdd_c.csv) used for intrusion detection.
- Contains 148,517 rows and 42 features (mixture of numerical + categorical attributes).
- Categorical features: protocol_type, service, flag, attack category.
- Numerical features: src_bytes, dst_bytes, duration, count, srv_count, etc.
- Attack categories include: benign, dos, probe, r2l, and u2r.
- LabelEncoder used to convert categorical attributes into numeric values.
- Dataset supports multi-class network intrusion detection task.

KNN MODEL OVERVIEW

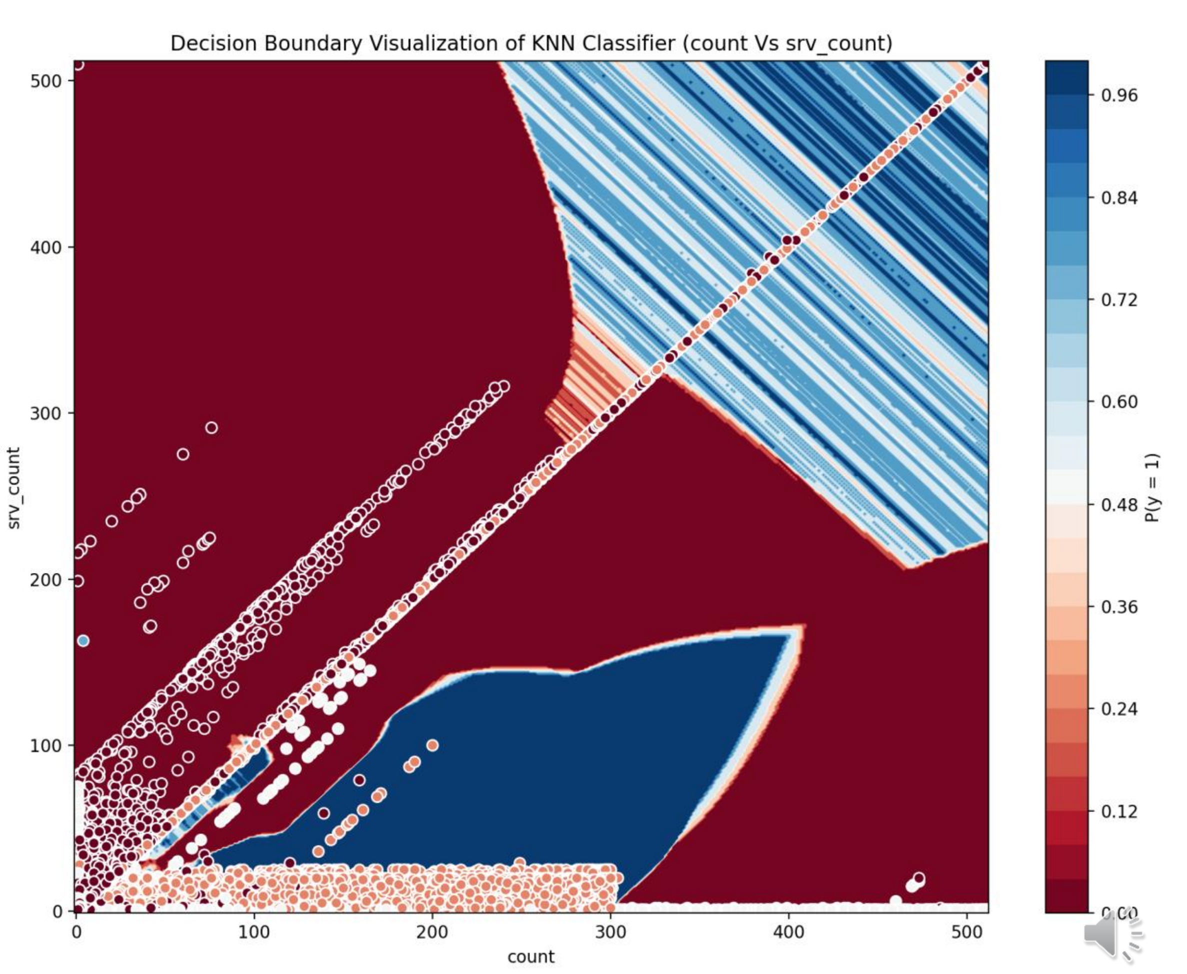
- KNN classifier (k=5) was trained on 100000 training samples.
- Tested on 35000 samples from the dataset..
- Model uses distance-based similarity to classify network traffic.
- Works well with multi-class problems such as intrusion detection.
- No training assumptions needed, relies on neighborhood structure in data.

DATA USED TO GENERATE THE PLOT

- Only two numeric features selected for visualization
  - count - number of connections to the same destination host
  - srv_count - number of connections to the same service
- These features show network-level behavior and allow meaningful separation.
- High-range features were avoided to prevent distortion in visualization.
- The boundary plot illustrates how KNN predicts probabilities in 2D feature space.

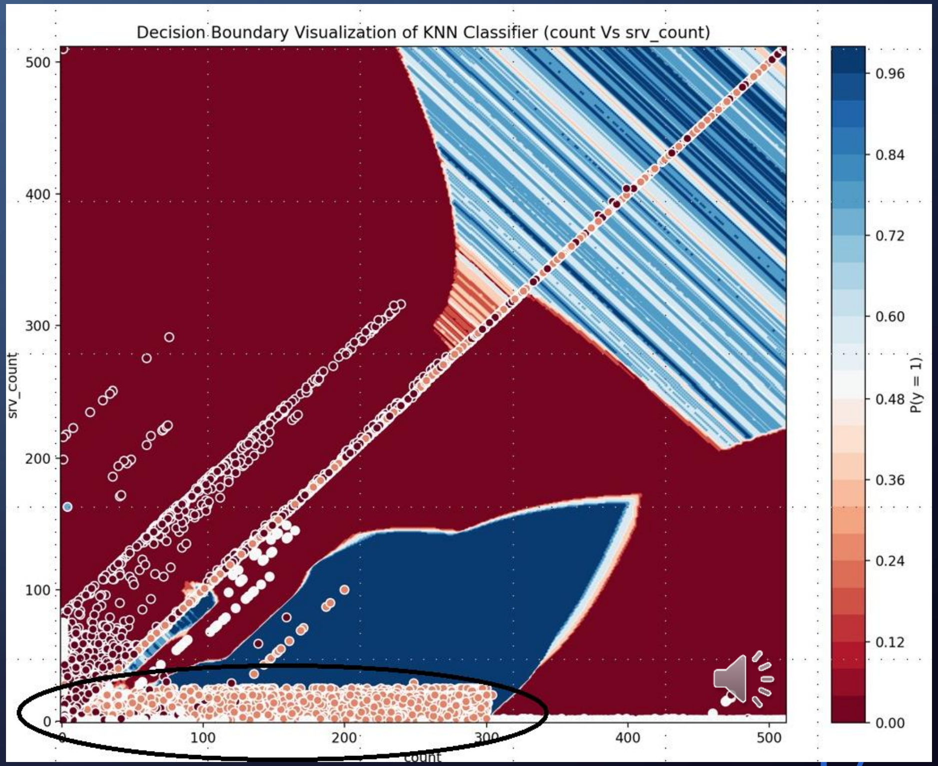
BOUNDARY PLOTS FOR THE TESTING THE MODEL

- Red regions = higher probability of attack.
- Blue regions = benign traffic
- White circular markers = actual test samples
- Benign traffic appears mostly in the lower-left region
- Attack traffic forms dense clusters along diagonal lines.
- The large red zones indicate broad areas where the classifier strongly predicts attacks.
- The blue central region suggests some separation between benign and attack traffic



IMPLICATIONS OF AN EVASION ATTACK

- Attacker can slightly change the data points (Red) to cross the decision boundary into a region the model classifies as "benign“ (Blue) to create an evasion attack.
- **Accuracy Degradation -**intended malicious instances can be labeled as benign
- **Increased False Negatives**
  - allow malicious activity to go unnoticed
- **Bypassed Security**
  - allow threat actor to access to the network
- **Operational Risk**
  - data breaches, system compromise



SUMMARY OF FINDINGS

- Accuracy achieved: **92.7%** and Error rate: **7.3%**
- Confusion matrix showed high performance for:
  - Benign: 17,552 correctly classified
  - DoS: 11,672 correct classifications
  - Probe: 2,876 correct, but some overlap with benign and DoS
  - R2L: 203 correct, still low because category is rare
  - U2R: 4 correct, very small sample size limits generalization
- Benign traffic appears mostly in the lower-left region where both count and srv_count are low.
- Attack traffic forms dense clusters along diagonal lines on the plot.
- Class imbalance affected performance, especially for rare attack types (R2L, U2R)

CONCLUSION

- The KNN classifier performed well on the KDD-based dataset and generated clear decision boundaries, achieving 92.7% accuracy with a 7.3% error rate.
- Preprocessing steps of encoding, large-scale sampling, and feature selection were essential to improve model accuracy and stability.
- Confusion matrix results showed strong performance for benign and DoS traffic, with moderate confusion on probe attacks due to overlapping statistical patterns.
- Rare classes such as R2L and U2R were detected but remain challenging due to their extremely low frequency, highlighting the impact of dataset imbalance.
- Overall, KNN is a solid baseline model for intrusion detection when properly tuned and supported with meaningful features.

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THANK YOU!